

5/21 - §8. Differential Entropy

Let X be random variable defined by

$X = x$ with probability $p(x)$

$$H(X) = - \sum_x p(x) \log p(x)$$

Let X be random variable

Def Cumulative distribution fn $F(x) := \Pr(X \leq x)$

Def X is continuous $\Leftrightarrow F$ is continuous

Def If $F'(x)$ exists, then $f(x) = F'(x)$ is called "probability density fn"

Def $h(X) := - \int_S f(x) \log f(x) dx$

$S := \{x \mid f(x) > 0\}$; support set of $\begin{pmatrix} f. \\ X. \end{pmatrix}$

Example $X \sim \mathcal{N}(0, \sigma^2)$, $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-x^2}{2\sigma^2}}$

$$h(X) = - \int_{-\infty}^{\infty} f(x) \log f(x) dx$$

$$= - \int_{-\infty}^{\infty} f(x) \left(\frac{-x^2}{2\sigma^2} \log e - \frac{1}{2} \log(2\pi\sigma^2) \right) dx$$

$$= \frac{\log e}{2\sigma^2} \text{var}(X) + \frac{1}{2} \log(2\pi\sigma^2)$$

$$= \frac{1}{2} \log e + \frac{1}{2} \log(2\pi\sigma^2) = \frac{1}{2} \log(2\pi e \sigma^2)$$

In ch. 3. AEP (By 문재영).

$X_1, \dots, X_n$ i.i.d $\sim p(x)$, $X \sim p(x)$

Thm 3.1.1 $-\frac{1}{n} \log p(X_1, \dots, X_n) \xrightarrow{\text{in probability}} H(X)$

$$A_\varepsilon^{(n)} := \{ (x_1, \dots, x_n) \mid 2^{-n(H+\varepsilon)} \leq p(x_1, \dots, x_n) \leq 2^{-n(H-\varepsilon)} \}$$

Thm 8.1.1 Let $X_1, \dots, X_n$ be i.i.d $\sim f(x)$

$$-\frac{1}{n} \log f(x_1, \dots, x_n) \xrightarrow{\text{in probability}} h(X)$$

$$(f(x_1, \dots, x_n) = \prod_{i=1}^n f(x_i))$$

Def $A_\varepsilon^{(n)} := \{ (x_1, \dots, x_n) : |-\frac{1}{n} \log f(x_1, \dots, x_n) - h(X)| \leq \varepsilon \}$

Thm 8.2.2 (Conti version of Thm 3.1.2)

1. $\Pr(A_\varepsilon^{(n)}) > 1 - \varepsilon$ for $n \gg 1$

2. $\text{Vol}(A_\varepsilon^{(n)}) \leq 2^{n(h+\varepsilon)}$ for all n

$$\text{Vol}(A) = \int_A dx_1 \dots dx_n$$

3. $\text{Vol}(A_\varepsilon^{(n)}) \geq (1 - \varepsilon) \cdot 2^{n(h-\varepsilon)}$ for $n \gg 1$

pf)

$$\begin{aligned} 2. \quad 1 &= \int_{S^n} f(x_1, \dots, x_n) dx_1 \dots dx_n \\ &\geq \int_{A_\varepsilon^{(n)}} 2^{-n(h+\varepsilon)} dx_1 \dots dx_n \\ &= 2^{-n(h+\varepsilon)} \text{Vol}(A_\varepsilon^{(n)}) \end{aligned}$$

$$\begin{aligned}
3. \quad 1 - \varepsilon &\leq \int_{A_\varepsilon^{(n)}} f(x_1, \dots, x_n) dx_1 \dots dx_n \\
&\leq \int_{A_\varepsilon^{(n)}} 2^{-n(h-\varepsilon)} dx_1 \dots dx_n \\
&= 2^{-n(h-\varepsilon)} \text{Vol}(A_\varepsilon^{(n)}) \quad (n \gg 1)
\end{aligned}$$

[Thm 8.2.3] (Exercise 3.11)

$A_\varepsilon^{(n)}$ is the smallest volume set with probability $\geq 1 - \varepsilon$ to first order in the exponent.

PP) Let $B_\varepsilon^{(n)}$ be the smallest volume set with probability $\geq 1 - \varepsilon$

$$\begin{aligned}
1 - 2\varepsilon &\leq \Pr(A_\varepsilon^{(n)} \cap B_\varepsilon^{(n)}) \\
&= \int_{A_\varepsilon^{(n)} \cap B_\varepsilon^{(n)}} f(x_1, \dots, x_n) dx_1 \dots dx_n \\
&\leq \int_{A_\varepsilon^{(n)} \cap B_\varepsilon^{(n)}} 2^{-n(h-\varepsilon)} dx_1 \dots dx_n \\
&\leq 2^{-n(h-\varepsilon)} \text{Vol}(B_\varepsilon^{(n)})
\end{aligned}$$

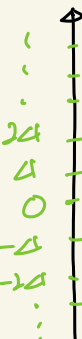
$$(1 - 2\varepsilon) 2^{n(h-\varepsilon)} \leq \text{Vol}(B_\varepsilon^{(n)})$$

$$(1 - \varepsilon) 2^{n(h-\varepsilon)} \leq \text{Vol}(A_\varepsilon^{(n)}) \leq 2^{n(h+\varepsilon)}$$

$$0 \leq \frac{1}{n} \log \frac{\text{Vol}(A_\varepsilon^{(n)})}{\text{Vol}(B_\varepsilon^{(n)})} \leq \frac{1}{n} \log \frac{2^{n(h+\varepsilon)}}{(1-2\varepsilon) 2^{n(h-\varepsilon)}} \xrightarrow{n \rightarrow \infty} 0$$

□

Digital data is quantized and has minimal unit Δ
 X ; conti r.v.



$$\Pr(i\Delta \leq X < (i+1)\Delta)$$

$$= \int_{i\Delta}^{(i+1)\Delta} f(x) dx = f(x_i) \cdot \Delta =: p_i$$

for some $i\Delta \leq x_i \leq (i+1)\Delta$

[Def] X^Δ ; quantized X

$X^\Delta = x_i$ with probability p_i

$$H(X^\Delta) = -\sum p_i \log p_i = -\sum f(x_i) \Delta \log f(x_i) - \sum f(x_i) \Delta \log \Delta$$

$$= -\sum f(x_i) \Delta \log f(x_i) - \log \Delta$$

$$H(X^\Delta) + \log \Delta = -\sum f(x_i) \Delta \log f(x_i)$$

$$\downarrow \Delta \rightarrow 0$$

$$- \int f(x) \log f(x) dx = h(X)$$

[Thm 8.3.1] $H(X^\Delta) + \log \Delta \xrightarrow{\Delta \rightarrow 0} h(X)$

[Example 8.3.1] $X \sim \text{Unif}[0,1]$, $\Delta = 2^{-n}$

$$\Rightarrow h(X) = 0, \quad H(X^\Delta) = n \xrightarrow{n \rightarrow \infty} \infty$$

In Ch 2. many concepts. like relative entropy, mutual information...

$$\boxed{\text{Def}} \quad h(x_1, \dots, x_n) = - \int f(x_1, \dots, x_n) \log f(x_1, \dots, x_n) dx_1 \dots dx_n$$

$\boxed{\text{Def}}$ (Conditional entropy)

$$\begin{aligned} H(X|Y) &= \sum_x p(x) H(Y|X=x) \\ &= - \sum_x p(x) \sum_y p(y|x) \log p(y|x) \\ &= - \sum_{x,y} p(x,y) \log p(y|x) \end{aligned}$$

$$\begin{aligned} \boxed{\text{Def}} \quad h(X|Y) &= - \int f(x,y) \log f(x|y) dx dy \\ &= - \int f(x,y) (\log f(x,y) - \log f(y)) dx dy \\ &= h(X,Y) - h(Y) \end{aligned}$$

$$\boxed{\text{Def}} \quad D(p||q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \quad (\text{relative entropy})$$

$$\boxed{\text{Def}} \quad D(f||g) = \int_S f \log \frac{f}{g}$$

$S := (\text{support set of } f) \subseteq (\text{support set of } g)$

$$\boxed{\text{Def}} \quad (\text{mutual information}) \quad I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$$

$$\boxed{\text{Def}} \quad I(X;Y) = \int f(x,y) \log \frac{f(x,y)}{f(x)f(y)} dx dy$$

[Thm 8.6.1] $D(f \| g) \geq 0$,
equality holds iff $f = g$ a.e.

$$\begin{aligned} \text{pf)} \quad -D(f \| g) &= \int_S f \log \frac{g}{f} \\ &\leq \log \int_S f \cdot \frac{g}{f} \leq \log \int_S g \leq 0 \end{aligned}$$

$$0 = D(f \| g) = \int_S f \log \frac{f}{g} \Leftrightarrow \frac{f}{g} = 1 \text{ a.e.}$$

[11]

[Cor]

$$\begin{aligned} I(X; Y) &= \int f(x, y) \log \frac{f(x, y)}{f(x)f(y)} dx dy \\ &= D(f(x, y) \| f(x)f(y)) \geq 0 \end{aligned}$$

equality holds $\Leftrightarrow f(x, y) = f(x)f(y)$ a.e

$\Leftrightarrow X$ and Y are independent.

[Cor]

$$\begin{aligned} I(X; Y) &= \int f(x, y) \log \frac{f(x, y)}{f(x)f(y)} dx dy \\ &= \int f(x, y) \log f(x|y) dx dy - \int f(x, y) \log f(x) dx dy \\ &= -h(X|Y) + h(X) \geq 0 \\ h(X|Y) &\leq h(X) \end{aligned}$$

$$h(x_1, \dots, x_n) = - \int f(x_1, \dots, x_n) \log f(x_1, \dots, x_n) dx_1 \dots dx_n$$

$$\begin{aligned} f(x_1, \dots, x_n) &= f(x_n | x_1, \dots, x_{n-1}) f(x_1, \dots, x_{n-1}) \\ &= f(x_n | x_1, \dots, x_{n-1}) f(x_{n-1} | x_1, \dots, x_{n-2}) f(x_1, \dots, x_{n-2}) \\ &= \dots = f(x_n | x_1, \dots, x_{n-1}) f(x_{n-1} | x_1, \dots, x_{n-2}) \dots f(x_1) \end{aligned}$$

$$h(x_1, \dots, x_n) = \sum_{i=1}^n h(x_i | x_1, \dots, x_{i-1}) \leq \sum_{i=1}^n h(x_i)$$

Example Let X be random vector $X \sim \mathcal{N}(0, K)$
 $\parallel$
 $(x_1, \dots, x_n)$

$$h(x_1, \dots, x_n) \leq \sum_{i=1}^n h(x_i) = \sum_{i=1}^n \frac{1}{2} \log(2\pi e K_{ii})$$

covariance matrix

|| Thm 8.4. ||

$$\frac{1}{2} \log(2\pi e)^n |K|$$

$$|K| \leq \prod_{i=1}^n K_{ii}$$

For all positive definite matrix K , we have

$$|K| \leq \prod_{i=1}^n K_{ii}$$

Hadamard's inequality

Thm 8-6.5) Let X be r.v. with $\begin{cases} \text{zero mean} \\ \text{covariance matrix } K \end{cases}$

$$h(X) \leq \frac{1}{2} \log(2\pi e)^n |K|$$

equality holds $\Leftrightarrow X \sim \mathcal{N}(0, K)$

pp) Let $\begin{cases} g \text{ be prob. density fcn of } X \\ \phi_K \text{ ---//--- of } \mathcal{N}(0, K) \end{cases}$

$$0 \leq D(g \parallel \phi_K) = \int g \log \frac{g}{\phi_K} = -h(X) - \int g \log \phi_K$$

$$\phi_K(x) = \frac{1}{(2\pi)^{\frac{n}{2}} |K|^{\frac{1}{2}}} e^{-\frac{1}{2} x^T K^{-1} x}$$

$\log \phi_K$ is quadratic

$$\int g(x) x_i x_j = \text{cov}(x_i, x_j) = K_{ij} = \int \phi_K(x) x_i x_j$$

$$\Rightarrow \int g \log \phi_K = \int \phi_K \log \phi_K$$

$$0 \leq -h(X) - \int \phi_K \log \phi_K = -h(X) + \frac{1}{2} \log(2\pi e)^n |K|$$

$$h(X) \leq \frac{1}{2} \log(2\pi e)^n |K|$$

equality holds $\Leftrightarrow g = \phi_K$ a.e

$\square$

Cor Let X be random variable with

$$EX = 0, \text{Var}(X) = \sigma$$

$$h(X) \leq \frac{1}{2} \log(2\pi e \sigma) = \frac{1}{2} \log(2\pi e \text{Var}(X))$$

$$\frac{1}{2\pi e} e^{2h(X)} \leq \text{Var}(X)$$